BEP proposal: what limits interception in a foosball-playing robot

Oct 12, 2026 · @Aarrav

Summary

This BEP asks what limits a foosball robot's ability to intercept the ball, and tests the answer on the Tech United table. The robot is the test bench; the measured answer is the thesis.

The table has an IR touch frame as ball sensor and one differential BLDC rod drive, not yet mounted. I will validate that one rod first, then the Tech United technician builds three more. Four rods can then run one simple strategy: the rod nearest the ball moves to it and kicks.

The deliverables are a measured latency budget, a comparison of three ball-state predictors, and block rate against ball speed on the real rod. A robot that beats human players stays the long-term goal.

Research question

What limits the interception performance of a foosball robot built on an IR touch frame and differential BLDC rod drives: sensing latency, prediction error, or actuator response? And how much does compensating the dominant limit improve the block rate as ball speed increases?

Four sub-questions break this down:

1. **Sensing.** What are the IR frame's report rate, latency, jitter, accuracy and dropout behaviour with a moving ball and men on the field?
2. **Prediction.** How much does delay-compensated prediction reduce the position error at the rod, for a constant-velocity model, a Kalman filter, and a Kalman filter with wall reflection?
3. **Actuation.** How fast can the differential drive position and rotate the rod, and how much of the remaining miss comes from actuator response?
4. **Decision.** Do the results show that a second sensor is needed?

Working hypothesis, to be tested: sensing latency dominates at high ball speed, the Kalman filter with wall reflection recovers most of it, and actuator response becomes the next limit.

The measures are end-to-end latency in ms split by stage, position error at the rod against prediction horizon and ball speed, and block rate against ball speed over repeatable shots.

State of the art and gap

In the sources reviewed so far, none reports an end-to-end latency budget for this kind of table or tests predictive interception against a reactive baseline. This is a first reading; the November literature review will confirm or correct it.

Work	Ball sensing	Left open for this BEP
Senden et al. 2022, IEEE RA-L	Overhead RGB camera, 100 Hz vision loop, self-calibrating world model	Tested on stationary ball positions; no latency or ball-speed figures; reactive alignment without prediction; no velocity in the world model; motion control not discussed
Real-time ball tracking in a semi-automated foosball table, RoboCup 2009	Overhead camera at 100 Hz, using prior knowledge of the table and rod positions	Tracking handles ball speeds up to 10 m/s with bounces and occlusion; the abstract reports no latency budget or interception rates

Other tables reach higher frame rates with the camera under a transparent field: 300 fps at EPFL and 500 fps in a XIMEA case study. This BEP keeps the existing IR frame and compensates its latency in software, as advised by the head of robotics.

Still to read: Janssen et al. 2010 and 2012 on this table, the KIcker deep-reinforcement-learning foosball system, and Jordy Senden's thesis.

Approach

The study runs in four steps, and each one produces a measurement.

1. **Characterize the IR frame.** Record raw HID reports and measure report rate, latency, jitter, linearity and dropouts, including with men on the field. Use an independent reference: two photogates at known positions for ball timing, and a scope or logic analyser to separate USB delay from sensing delay. Host timestamps alone are not trusted.
2. **Build the latency budget.** Split the delay into sensor and USB, compute, the EtherCAT command path, and actuator response. The actuator part comes from bench step responses.
3. **Compare predictors offline.** Replay recorded ball rolls through three predictors: constant velocity, a Kalman filter on ball position and velocity, and the same filter with wall reflection. Reflection mirrors the lateral position at the walls with a restitution factor fitted from the recordings. Bounces off men and rods cannot be predicted, so a jump in the filter's innovation inflates its uncertainty or resets the velocity.
4. **Test closed loop.** On the mounted rod, fire repeatable shots with measured speed, using a ramp or spring launcher and photogates. Measure block rate against ball speed for the reactive baseline and for each predictor.

Delay compensation propagates the filter state forward by the measured latency. The rod target then follows from the predicted state, with the ball at (x, y), velocity (vx, vy) and rod position xr:

```
t = \frac{x_r - x}{v_x}, \qquad y_{target} = y + v_y \, t
```

The differential drive couples translation and rotation, because each motor contributes to both. Actuator work therefore covers decoupled control, fast positioning under torque and speed limits, and kick speed. A small 2D simulator (point ball, walls, actuator model) tunes the predictors and tests the four-rod strategy before the hardware exists.

Requirements and scope

The BEP is complete when Level 1 is measured; Levels 2 and 3 extend it if time allows.

Level	What the robot does	Evidence
1 (required)	The goalie rod blocks shots, with the reactive baseline and with prediction	Block rate against ball speed, latency budget, predictor comparison

Level	What the robot does	Evidence
2	Four rods run the same controller: the rod nearest the ball moves to it and kicks	The same measurements on more rods, in simulation and on the table
3 (stretch)	A play session against human players	Goals and blocks logged; reported as a demo, not required

Requirements:

- **R1.** Latency is measured with an independent reference, not host timestamps.
- **R2.** The predictor runs in real time at the IR frame's report rate.
- **R3.** Shots are repeatable, with speed measured by photogates.
- **R4.** One controller implementation is instantiated per rod, so adding a rod needs configuration, not new code.
- **R5.** Numeric targets (shots per speed, acceptable prediction error, required prediction horizon) are set at milestone M1, once the latency budget is known.

Out of scope: passing, aimed kicks to a setpoint, mechanical robustness and portability, and a new camera or event camera. A second sensor is added only if the results show the IR frame is a serious limit.

Work packages, milestones and gate

Six work packages fill the 280 BEP hours; work done before 1 February comes on top and is unofficial. Hours are a draft estimate to review with the supervisors.

Work package	Work	Output	Draft hours
WP0 Pre-work (Nov to Jan)	Literature review, IR frame characterization, proposal, mount the first rod, EtherCAT bring-up	Characterization results, mounted rod, decision at gate G1	Not counted
WP1 Latency budget and baseline	Measure end-to-end latency by stage; reactive controller on the rod	Latency budget, baseline block rate, numeric targets	30

Work package	Work	Output	Draft hours
WP2 Prediction study	Offline comparison of three predictors with delay compensation	Prediction error against horizon and ball speed	60
WP3 Actuator control	Decoupled differential control, fast positioning, kick speed	Step responses, positioning time, kick speeds	50
WP4 Closed-loop experiments	Block rate against ball speed for baseline and predictors	Results on the first rod, second-sensor decision	50
WP5 Four rods and play session (stretch)	The same controller on four rods; a play session against humans	Evidence for Levels 2 and 3	20
WP6 Paper and presentation	Analysis, paper, panel talk	Final paper and presentation	70

The first rod is the critical path. The technician has offered to procure and manufacture three more setups once one has been tested and works well. Every week that gate G1 slips moves Level 2 later.

Gate or milestone	Target	Done when
G1 First rod validated	15 Jan 2027	Rod mounted on the table; both motors driven through the TU/e EtherCAT drivers; translation and rotation commanded independently; step response recorded on both axes; a kick test completes without damage. The technician is then asked to start on three more setups.
M1 Baseline	12 Feb 2027	Latency budget and baseline block rate measured; numeric targets set
M2 Predictors compared	5 Mar 2027	Offline comparison finished, required prediction horizon known
M3 Rod 1 results	16 Apr 2027	Block rate against ball speed measured; second-sensor decision made

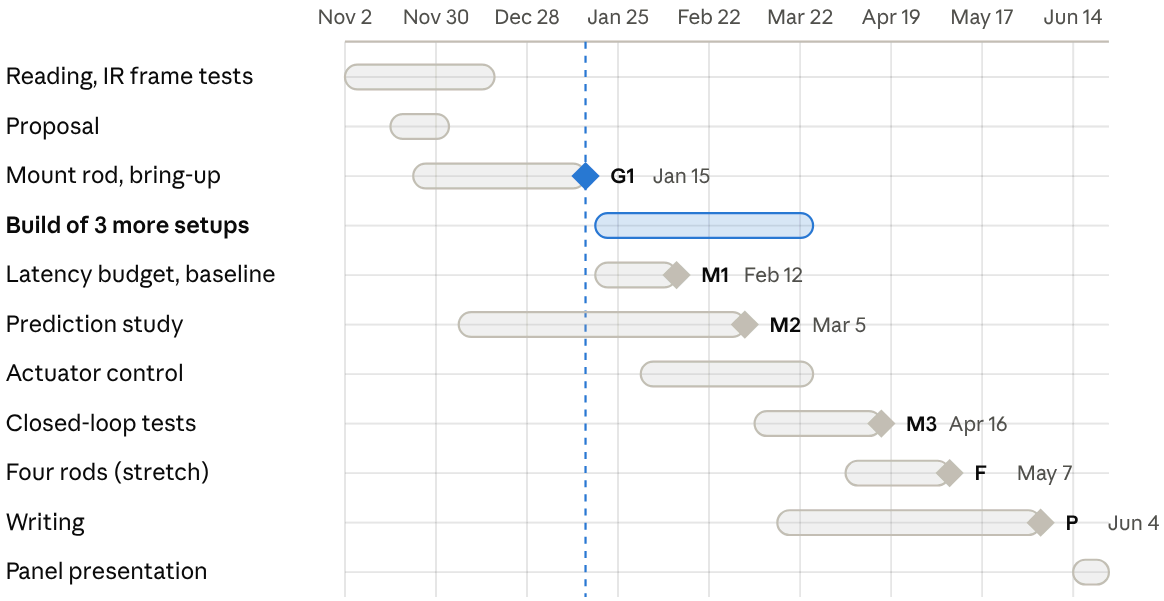
Gate or milestone	Target	Done when
F Hardware freeze	7 May 2027	No hardware changes after this date

The G1 criteria are proposed and still to be agreed with the technician and the daily supervisor. The lead time for procuring and building three setups is not yet known; the timeline assumes about ten weeks.

Timeline

The other three rods wait on gate G1 in mid-January

BEP plan, November 2026 to June 2027; all dates are targets



G1 first rod validated; M1 latency baseline; M2 predictors compared
M3 rod 1 results done; F hardware freeze; P paper due (assumed date)

Gantt chart · Nov 2026 to Jun 2027, 11 lanes, gate G1 and five markers

The first rod must pass gate G1 on 15 January before the technician starts on three more setups. Dates assume a 1 February start and an early-June paper deadline, and exam weeks are not yet marked.

Risks and mitigation

The biggest risk is a late gate G1, because everything beyond Level 1 depends on it.

Risk	Effect	Mitigation
Gate G1 slips	The three extra setups start late and Level 2 is not reached	Start mounting in November; agree the G1 criteria now; confirm the technician's lead time; Level 1 needs only one rod
IR frame gives ghost points or dropouts with men on the field	Predictions are unreliable	Characterize it early in WP0; reject outliers using the filter's innovation; add a second sensor only if the results demand it
Shot speed varies between launches	Block rate is hard to compare across speeds	Measure every shot with the photogates and group shots by measured speed
Sensing is not the dominant limit	The working hypothesis is wrong	The latency budget names whichever stage dominates, and the improvement study then targets that stage
The goal drifts towards a full playing robot	Thin research results	Staged levels; Level 1 defines completion; hardware freeze on 7 May
June overlaps with the honors project final demo	Less time in June	No hardware work after the freeze; paper drafting starts in March

Resources, supervision and open points

The project reuses Tech United's table and existing hardware, and needs the technician's workshop capacity for three more setups.

- **Hardware:** the foosball table with its IR touch frame, one differential rod drive (two Turnigy 6374-149KV BLDC motors, timing belts, bevel gears), and the TU/e EtherCAT motor drivers. To add: photogates, a shot launcher, and a scope or logic analyser.

- **Supervision:** Jake Rap as daily supervisor; an EE professor as supervisor, approached with Jake's help. Contacts: Elise for the actuators, Jordy Senden for perception and the world model.
- **Time:** 10 credits (280 h), with unofficial work from November and the formal start in February 2027.

Open points:

- [ ] Confirm the private-proposal deadline, paper deadline and presentation window on the EE BEP Canvas page
- [ ] Get the technician's lead time for three more setups
- [ ] Find the EE supervisor
- [ ] Find out what control software exists for the TU/e EtherCAT drivers
- [ ] Agree the G1 criteria with the technician and Jake
- [ ] Decide whether an ME student joins as a combined BEP, which Jordy raised as an option